

SORCS1 knockout gRNA summary

SpCas9 · NHEJ frameshift · Homo sapiens · RefSeqGene NG_029120.2
MANE transcript NM_052918.5 / ENST00000263054 · Generated 2026-08-12

STRATEGY

Dual SpCas9 guide RNAs targeting coding **exon 4** to induce NHEJ indels and frameshift knockout. Early constitutive exons are preferred; extreme N- and C-terminal exons are avoided to reduce alternative start usage and residual protein risk.

GENE

SORCS1

SPECIES

Homo sapiens

TARGET EXON

Exon 4 (159 bp coding)

NUCLEASE / PAM

SpCas9 · NGG

RECOMMENDED PAIR (DESIGNER AUTO-SELECT)

Source: ASSURED designer on NG_029120.2 · Cut spacing 103 bp

GUIDE	SPACER (20 NT, ORDER WITHOUT PAM)	PAM	STRAND	GC%
SORCS1_KO_gRNA1	ATAAACTGCTCTCAATCTCC	GGG	–	40
SORCS1_KO_gRNA2	TTCACCCCAAACAAGAAGAC	TGG	+	45

SCREENING PRIMERS (RECOMMENDED PAIR)

PRIMER	SEQUENCE	AMPLICON
SORCS1_KO_Fw	AAGCATTCTTTGCCTGCC	~450 bp
SORCS1_KO_Rev	TTTAAACAAGCAACCTGAGACTC	

VALIDATED HISTORICAL EXON-4 PAIR

Historical hiPSC KO projects (SorCS1-Exn4-gRNA1/2); established lines MDCi053-A-27 / HMGUi001-A-28 · Spacing 7 bp

GUIDE	SPACER (20 NT)	PAM	STRAND
SorCS1-Exn4-gRNA1	GATAATGTTACTCACAGACC	CGG	+
SorCS1-Exn4-gRNA2	ATAAACTGCTCTCAATCTCC	GGG	-

Primers: TCAAGGTCCTGTTTGGC / AGACTCTCCTTGCTTTCCC (~452 bp)

BRUNELLO LIBRARY GUIDES (HUMAN)

SPACER	PAM	EXON	RULE SET 2
AGAACAGGGGACGCACTACG	AGG	1	0.6693
AGCTCTATGACTATAACCTG	GGG	2	0.7234
ATGTGTGTTTCATAGCAACC	AGG	13	0.6807
CCTCGTGACATACCTCATAG	AGG	9	0.6690

MOUSE ORTHOLOG CANDIDATES (NM_021377)

SPACER	PAM	EXON
GTTGATCAGCTCAGATGAAG	GGG	4
GATTATGTTACTCACAGACC	CGG	4
GCAGCTTATCCAGGAATCAG	TGG	5

ORDERING & VALIDATION NOTES

- Order spacers **without** PAM; use SpCas9 (NGG).
- For U6 expression, prepend a leading G if the spacer does not start with G.
- Sanger-sequence the target locus in your parental line before ordering (SNPs can block cutting).
- Test 2–4 guides; dual-guide RNP increases indel rate.
- Validate KO by ICE/TIDE plus protein loss (western/IF); prefer frameshifting indels.

Files: outputs/sorcs1_knockout_design_card.json, outputs/sorcs1_ko_auto.json, outputs/sorcs1_ko_validated.json, references/sorcs1-ng_029120.2.gb